SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

COMPUTER SCIENCE AND ENGINEERING PROGRAMME

CSA3717 – SOFTWARE TESTING

EXPERIMENT 11

RAILWAY RESERVATION SYSTEM – POSITIVE TEST SCENARIOS AND TEST CASES

Aim:

To Write a case study to identify and analyze the requirements of a Railway Reservation System and prepare positive test scenarios and test cases for the application.

1. Problem Description

A railway reservation system allows passengers to search trains, check seat availability, enter passenger details, select a class, pay reservation charges and receive a booking confirmation or ticket.

The Railway Reservation System is expected to provide reliable, secure and user-friendly services. The testing team verifies the stated requirements using valid and invalid inputs, records actual results, and reports any deviation from the expected behaviour.

2. Functional Requirements

1. Passengers must be able to search trains using source, destination and journey date.
2. The system must show train schedules and class-wise availability.
3. Passenger details must be validated before booking.
4. Seat allocation must not exceed available capacity.
5. Fare must be calculated according to route, class and applicable charges.
6. Booking is confirmed only after successful payment and seat allocation.
7. A confirmed booking must generate a unique PNR/ticket reference.
8. Users must be able to view or cancel bookings according to policy.

3. Positive Test Scenarios and Test Cases

Positive testing checks whether the application performs the required operation when valid data and correct user actions are supplied.

Test Case ID	Test Scenario	Test Input / Action	Expected Result	Status
PT01	Search trains	Enter valid source, destination and future journey date	Matching trains are displayed.	Pass
PT02	Check availability	Select a train and class	Current seat availability is displayed.	Pass
PT03	Enter passenger details	Provide valid passenger name, age and required details	Passenger details are accepted.	Pass
PT04	Select class	Choose a class with available seats	Class selection is accepted.	Pass
PT05	Calculate fare	Request fare for selected route and class	Correct fare breakdown is displayed.	Pass
PT06	Make payment	Complete payment successfully	Payment is successful.	Pass
PT07	Confirm booking	Complete booking with available seat	Unique PNR and ticket are generated.	Pass
PT08	View booking	Search using valid PNR	Correct booking details are displayed.	Pass

5. Detailed Execution of a Failed Test Case

Test Case ID: PT08

Precondition: A registered user has a valid account and the required service is available.

Test Steps: 1. Sign in with valid credentials. 2. Open the relevant module. 3. Enter valid data. 4. Submit the operation.

Expected Result: The application completes the operation and displays a success confirmation.

Actual Result: The operation completes successfully in this illustrative execution.

Conclusion: The positive test passes when actual behaviour matches the expected result.

Conclusion

The test cases prepared for the Railway Reservation System provide coverage of its key functional requirements. Executing them helps identify functional gaps, improve error handling and increase confidence in the quality of the application.